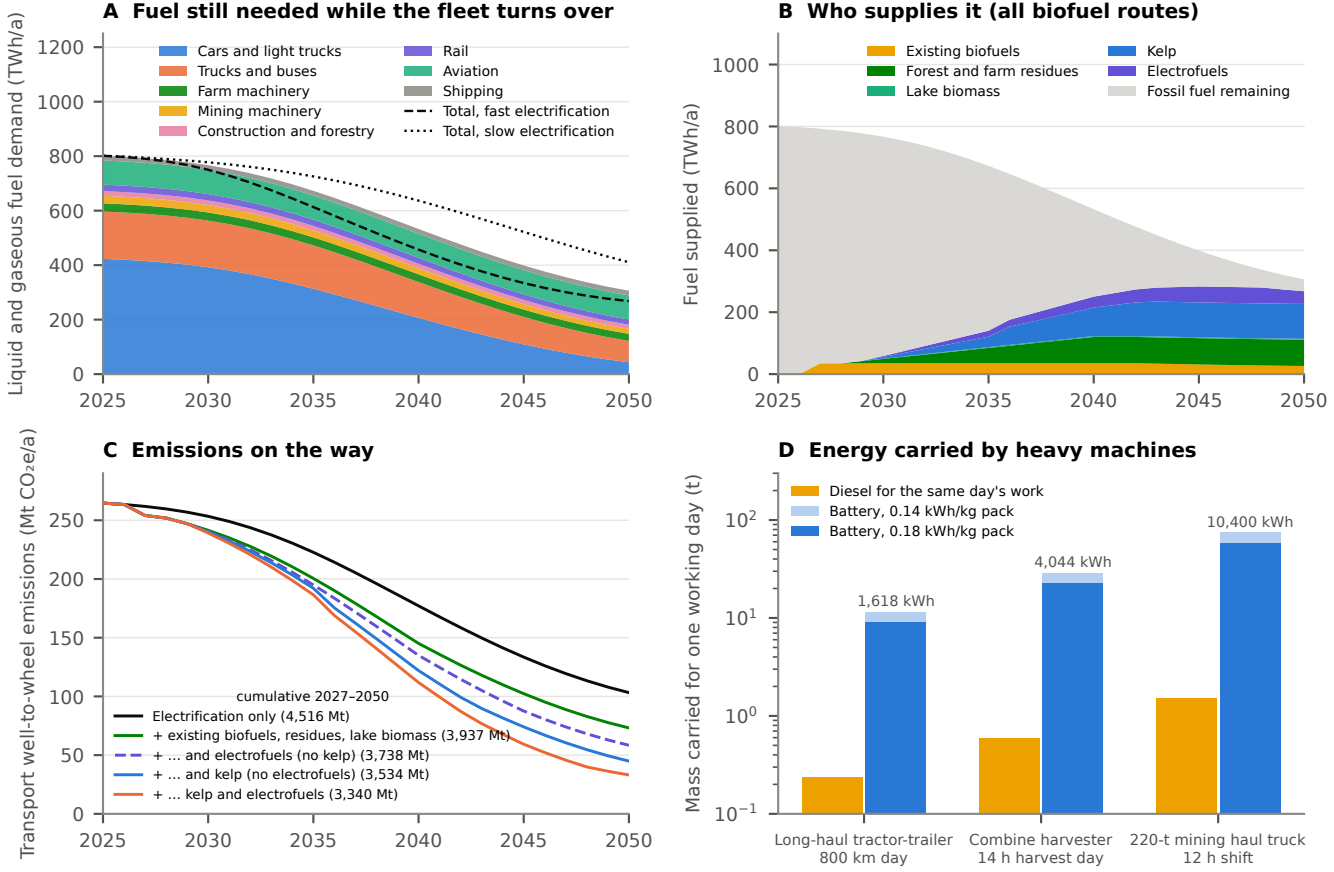
A Fuel still needed while the fleet turns over
Liquid and gaseous fuel demand (TWh/a)
Cars and light trucks
Trucks and buses
Farm machinery
Mining machinery
Construction and forestry
Rail
Aviation
Shipping
Total, fast electrification
Total, slow electrification
B Who supplies it (all biofuel routes)
Fuel supplied (TWh/a)
Existing biofuels
Forest and farm residues
Lake biomass
Kelp
Electrofuels
Fossil fuel remaining
C Emissions on the way
Transport well-to-wheel emissions (Mt CO₂e/a)
cumulative 2027–2050
Electrification only (4,516 Mt)
+ existing biofuels, residues, lake biomass (3,937 Mt)
+ ... and electrofuels (no kelp) (3,738 Mt)
+ ... and kelp (no electrofuels) (3,534 Mt)
+ ... kelp and electrofuels (3,340 Mt)
D Energy carried by heavy machines
Mass carried for one working day (t)
Diesel for the same day's work
Battery, 0.14 kWh/kg pack
Battery, 0.18 kWh/kg pack
1,618 kWh
4,044 kWh
10,400 kWh
Long-haul tractor-trailer
800 km day
Combine harvester
14 h harvest day
220-t mining haul truck
12 h shift